

VAISHNAVI PUNDE

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Summary

Data Science student with strong foundations in machine learning, deep learning, and data analytics. Experienced in building end-to-end ML pipelines, performing data preprocessing, and developing predictive models using real-world datasets. Proficient in Python, SQL, and Power BI for extracting insights and creating interactive dashboards. Passionate about applying AI solutions to solve practical problems across healthcare, analytics, and intelligent systems.

Skills

Data Analysis & AI: Python (Pandas, NumPy, Scikit-learn), SQL.

Data Visualization Tools: Power BI, Matplotlib

Databases: MySQL, PostgreSQL, SQLite.

Tools & Version Control: Git, GitHub, Jupyter Notebooks, VS Code, Claude.

Projects

Multi-Format Document Redaction System | Python, FastAPI, React, NLP, Presidio, spaCy, PyMuPDF, OCR, SQLite

- Built a full-stack intelligent web application that automates detection and redaction of Personally Identifiable Information (PII) across PDF, image (PNG/JPEG), Excel, and CSV document formats.
- Implemented a multi-layered PII detection engine combining Microsoft Presidio with spaCy NER models and custom regex patterns for 15+ PII categories including Indian identifiers (Aadhaar, PAN, Passport), achieving an overall F1-score of 95.6%.
- Integrated Tesseract OCR via pdf2image for scanned document processing, and designed a RESTful FastAPI backend supporting single-document and batch processing with an SQLite-backed audit trail for compliance.
- Developed an interactive React + Vite frontend with real-time PII highlights, configurable redaction styles (blackbar, text label, blur, strikethrough), false-positive review workflow, and redaction profile management.

Medicine Suggestion System | Python, Django, HTML, CSS, SQLite

- Developed a Django-based web application that suggests medicines based on user-entered symptoms.
- Implemented secure form handling using CSRF protection and backend logic for symptom processing.
- Designed a responsive and user-friendly interface using HTML and CSS.
- Applied MVC architecture to ensure clear separation of backend logic and frontend presentation.

AI-Powered Personal Expense Tracker | Python, Django, Pandas, Matplotlib, SQLite

- Developed a full-stack web application using Django to help users track and manage their daily financial transactions.
- Implemented automated data categorization using Python (Pandas) to sort expenses into groups like Food, Travel, and Bills.
- Created a dynamic dashboard with Matplotlib to visualize spending trends and monthly budget distributions.
- Utilized SQLite to manage user data and transaction history efficiently while following the MVC architecture.

Iris-Based Person Identification System | Python, CNN, Siamese Network, OpenCV, ORB, LBP, Streamlit

- Developed a biometric identification system using iris images for secure person verification.
- Implemented a CNN-based Siamese Network to compare iris image pairs using similarity learning.
- Enhanced accuracy by integrating ORB for structural features and LBP for texture feature extraction.
- Deployed the system using Streamlit with image preprocessing and balanced dataset handling.

Awards & Certifications

Guinness World Record for the Largest Online Photo Album of Books (December 2024 — 160 hours): Contributed to a Guinness World Record project by performing data collection, cleaning, preprocessing, coordination, and management of large-scale datasets during a 160-hour internship at Savitribai Phule Pune University.

Education

B.Sc. Data Science

Graduated: 2026

Department of Technology, SPPU, Pune

CGPA: 9.69 / 10

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Statistics, Data Analytics, Computer Vision, Natural Language Processing